

Count Occurrences in Seven Integers Using Java Single Dimension Arrays

In this assessment, you will design and code a Java console application that reads in seven integer values and prints out the number of occurrences of each value. The application uses the Java single-dimension array construct to implement its functionality. You can use either the Toolwire environment or your local Java development environment to complete this assignment.

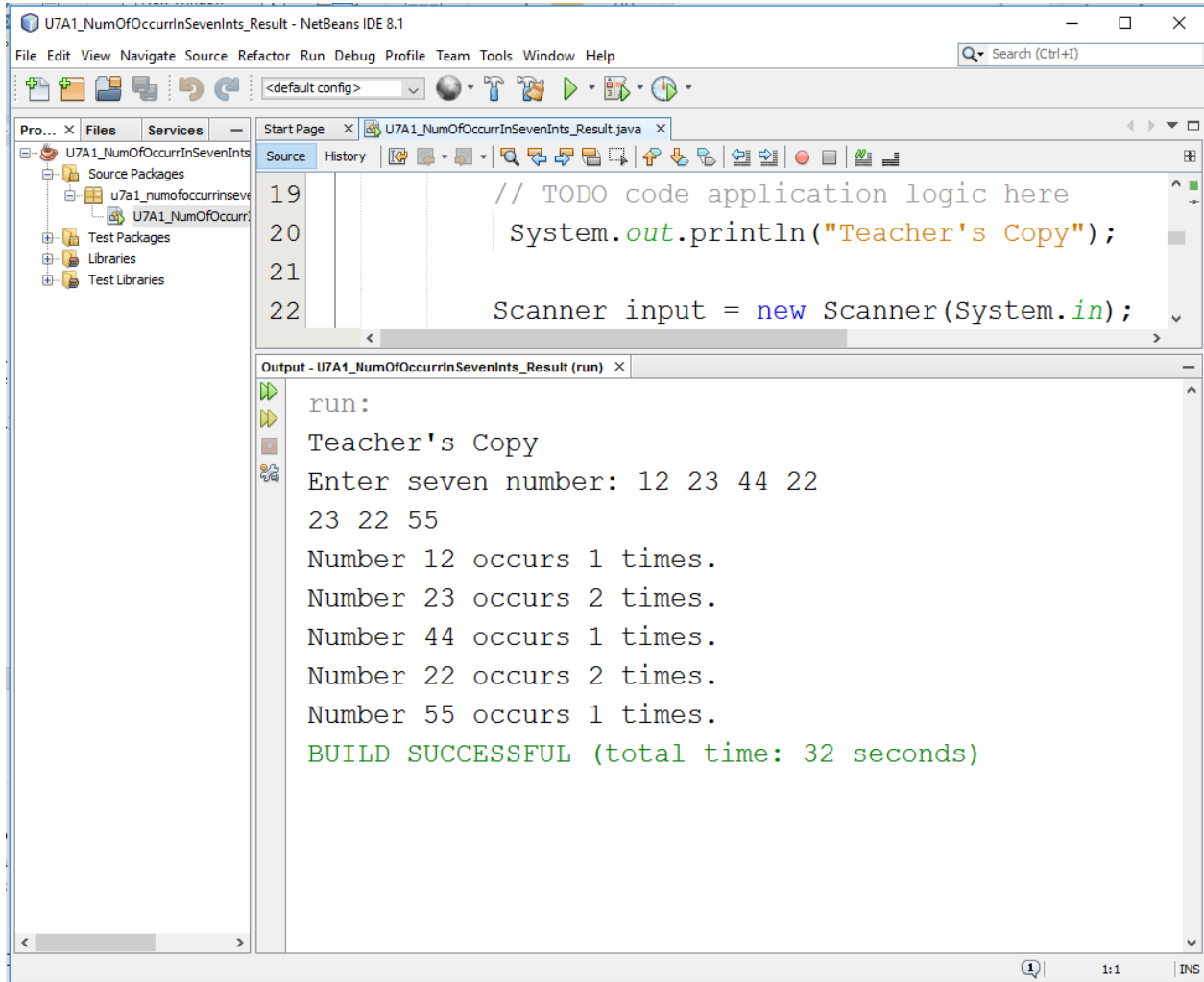
The requirements of this application are as follows: The application is to read seven integer numbers entered by the user from the keyboard. Using Java single-dimension arrays, the application counts the number of occurrences of each of the seven values. The application then prints out the number of occurrences of each of the seven values to the screen.

Assume that the user always entered correct information so there is no need to validate user input in your application.

Use these three input values to test your application:

12 23 44 22 23 22 55.

Successful completion of this assignment will show the correct occurrence count of each of the seven entered integers when the application is run. Your program output should look like this sample output:



```
U7A1_NumOfOccurrInSevenInts_Result - NetBeans IDE 8.1
File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help
Search (Ctrl+F)

Pro... x Files Services
U7A1_NumOfOccurrInSevenInts
  Source Packages
    u7a1_numofoccurrinseve
      U7A1_NumOfOccurrInSevenInts_Result.java
  Test Packages
  Libraries
  Test Libraries

Start Page x U7A1_NumOfOccurrInSevenInts_Result.java x
Source History
19 // TODO code application logic here
20 System.out.println("Teacher's Copy");
21
22 Scanner input = new Scanner(System.in);

Output - U7A1_NumOfOccurrInSevenInts_Result (run) x
run:
Teacher's Copy
Enter seven number: 12 23 44 22
23 22 55
Number 12 occurs 1 times.
Number 23 occurs 2 times.
Number 44 occurs 1 times.
Number 22 occurs 2 times.
Number 55 occurs 1 times.
BUILD SUCCESSFUL (total time: 32 seconds)
```

Follow these steps to complete this assignment:

1. Create a NetBeans Java console application project called "U7A1_NumOfOccurrInSevenInts".
2. Develop the application to meet the stated requirements.
3. Compile and test your application using the provided input data.

Make sure to document the result of your testing by taking screenshots of the result of running your application similar to the provided sample output. One screenshot is required for this assignment.

4. Explain the approach you took to complete this assignment and the major decisions you made. As part of your explanation, be sure to identify the fundamental Java constructs you used that were specific and relevant to your submitted program.

Deliverables:

Use the submission template available in the resources (WeekXSolutionSubmissionTemplate.docx) to complete and submit your deliverables. Your deliverables in the attached submission template should include:

1. Your work (Netbeans project zip file + copy of *.java source code)
2. A screenshot of the result of testing your application. See the examples above.
3. Explain the approach you took to complete this assignment and the major decisions you made. As part of your explanation, be sure to identify the fundamental Java constructs you used that were specific and relevant to your submitted program.

Your assignment will be scored on the following criteria:

1. Design a program that meets given specifications.
2. Code an application that exercises single array constructs.
3. Test the application and document testing.
4. Identify relevant fundamental constructs in submitted program.
5. Explain the approach taken to develop the application and the major decisions made.
6. Communicate efficiently, effectively, and in an appropriate manner as an IT professional.